

Software Requirement Specification (SRS)

ShopEasy – E-Commerce Management Database System

1. Introduction

The ShopEasy – E-Commerce Management Database System is designed to manage the core operations of an online shopping platform through a centralized relational database. The system stores and organizes customer information, products, categories, suppliers, inventory, orders, payments, shipments, and customer reviews. It helps the business maintain accurate records, reduce data redundancy, and improve operational efficiency.

2. Purpose

The purpose of this Software Requirement Specification (SRS) is to define the functional and non-functional requirements of the ShopEasy – E-Commerce Management Database System. This document serves as a guideline for database design, ER modeling, normalization, SQL implementation, testing, and future system maintenance.

3. Scope

The system will provide the following functionalities:

- Customer registration and profile management.
- Product and category management.
- Supplier information management.
- Inventory and stock monitoring.
- Order placement and tracking.
- Payment recording and status management.
- Shipment and delivery tracking.
- Customer review and rating management.
- Business report generation.
- Centralized storage and retrieval of business data.

4. Overall System Description

The system maintains all e-commerce information in a centralized database using the following entities:

- Customer
- Category
- Product
- Supplier
- Order
- Order Details
- Payment
- Shipment
- Review

Authorized users can create, update, retrieve, and manage records while ensuring data consistency and integrity.

5. Functional Requirements

5.1 Customer Management

- Register new customers.
- Store customer details such as name, email, phone number, and address.
- Update customer information when required.

5.2 Category Management

- Create and maintain product categories.
- Assign products to appropriate categories.

5.3 Product Management

- Add new products.
- Update product information.
- Maintain product prices, descriptions, and stock quantities.

5.4 Supplier Management

- Store supplier details.
- Maintain supplier contact information.

5.5 Inventory Management

- Monitor available stock.
- Update stock quantities after each order.
- Identify low-stock products.

5.6 Order Management

- Create customer orders.
- Store order details.
- Maintain order status (Pending, Processing, Delivered, Cancelled).
- Record ordered products with quantity and unit price.

5.7 Payment Management

- Record payment details.
- Maintain payment status.
- Store payment method and payment date.

5.8 Shipment Management

- Record shipment information.

- Maintain delivery address.
- Update shipment and delivery status.

5.9 Review Management

- Store customer ratings.
- Record customer comments for products.

5.10 Report Generation

- Customer reports.
- Product reports.
- Order reports.
- Payment reports.
- Shipment reports.
- Monthly sales reports.

6. Non-Functional Requirements

Security

- Only authorized users can access the database.
- Customer and business data must be protected.

Reliability

- Provide consistent and accurate data processing.

Performance

- Retrieve and update records with minimal response time.

Availability

- Database should be available during business operations.

Data Integrity

- Maintain accurate and consistent data.

Scalability

- Support future business growth.

Maintainability

- Allow future modifications with minimal effort.

Backup & Recovery

- Protect data through backup and recovery mechanisms.

Usability

- Provide a simple and user-friendly interface.

Compatibility

- Work with standard DBMS platforms.

7. Assumptions

- Customers provide accurate registration details.
- Products are categorized correctly.
- Inventory is updated after every order.
- Each order belongs to one registered customer.
- Payment and shipment information is entered correctly.
- Reviews are submitted only by registered customers.

8. Constraints

- Only predefined entities and attributes are used.
- Entity names and attributes cannot be modified.
- The project focuses only on the database system.
- Online payment gateway integration is outside the scope.
- Internet connectivity is required.
- Data accuracy depends on valid user input.

9. Conclusion

The ShopEasy – E-Commerce Management Database System provides a secure, reliable, and centralized platform for managing e-commerce operations. This SRS establishes a clear foundation for database design, implementation, testing, and future enhancements. By fulfilling the specified requirements, the system will improve data organization, streamline order processing, enhance inventory control, and support efficient business decision-making.